CFD Analysis of a Geothermal Steam-Water Separator: Spiral-Inlet Replication, DPM Tracking and Wall-Film Sensitivity

Technical results from controlled ANSYS Fluent simulations and automated PyFluent post-processing

1. Numerical model and automated workflow

The study used staged CFD comparisons to quantify liquid carryover in a vertical bottom-outlet cyclone separator. The first model family reproduced six operating conditions from Purnanto et al. (2013) on the available rectangular 90° spiral-inlet geometry. A second family retained the separator geometry while introducing split steam-liquid inlet and Eulerian wall-film (EWF) treatments to isolate model-form sensitivity. Results are therefore separated into reference-replication evidence and exploratory wall-interaction evidence.

Table 1. Principal numerical settings and controlled model differences.

Item	Spiral-inlet replication	Split-inlet / EWF family
Mesh	2,964,593 tetrahedral cells	7,601,261 cells
Solver	Steady, pressure based; Mixture model	Steady, pressure based; Mixture + DPM/EWF
Turbulence	RNG k-ε; standard wall functions	RNG k-ε; standard wall functions
Thermal scope	Energy disabled; no flashing	Energy disabled; no flashing
Inlet	Single two-phase mass-flow inlet	Separate steam and liquid mass-flow regions
Outlet / lower boundary	1.12 MPa pressure outlet / DPM trap	Pressure outlet / closed lower wall
Discrete phase	Nine Harwell-derived bins; one-way DPM	Deterministic/stochastic DPM and EWF branches

1.1. Operating-condition and injection reconstruction

Each enthalpy label selected the steam-liquid mass split reported by Purnanto; enthalpy was not solved as a thermal boundary condition. For each case, the Harwell correlation generated a dimensional droplet distribution from nine fixed normalized diameters. The spiral baseline's native surface-injection convention was preserved: inert liquid-water particles were released from the inlet using the local face-normal direction. Consequently, the spiral inlet's physical orientation was respected without imposing a Bangma-specific global velocity vector.

$$x_{sa} = 1.91\, D_t\, (Re^{0.1} / We^{0.6})(\rho g / \rho l)^{0.6}, \quad x_{me}d = 1.42\, x_{sa}$$

The automated sequence loaded a fresh baseline, set and read back both phase mass flows, updated all nine injections, hybrid-initialised the carrier flow, completed 1500 verified iterations in 25-iteration blocks, saved a pre-DPM state, performed DPM tracking, and exported injection-level fates. Early residual stopping was disabled. A case was accepted only when all 54 injection rows were populated and escaped, trapped and incomplete mass reconciled with injected liquid within 0.2%.

$$\dot{m}inj \approx \dot{m}escaped + \dot{m}trapped + \dot{m}incomplete, \quad quality = \dot{m}steam /(\dot{m}steam + \dot{m}escaped) \times 100\%$$

2. Completed spiral-inlet enthalpy sweep

All six operating conditions completed 1500 carrier-flow iterations on 15 Fluent ranks, followed by per-injection DPM tracking. Escaped liquid remained between 0.01655 and 0.02667 kg/s, giving calculated outlet steam quality between 99.96681% and 99.97949%. The small absolute range is technically important: the present CFD points form a nearly flat band and do not reproduce either the pronounced 26.85 m/s dip in Purnanto's digitised simulation series or the high-velocity decline in its calculation series.

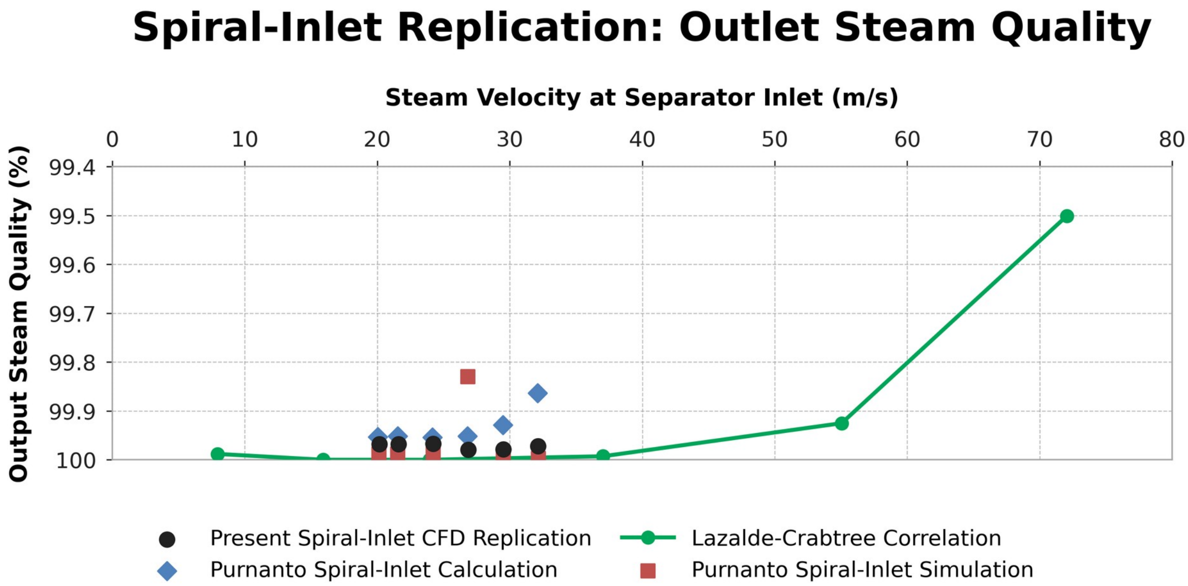

Figure 1. Present six-case spiral-inlet CFD results compared with digitised Purnanto calculation/simulation points and the Lazalde-Crabtree correlation. The quality axis is inverted to match the source figure.

Table 2. Completed spiral-inlet DPM outcomes and comparison with the nearest digitised simulation point.

Case	Condition	v (m/s)	Escaped (kg/s)	Incomplete (kg/s)	Quality (%)	Δ vs ref. (pp)
1	1600 -25%	20.15	0.01941	2.068	99.9679	-0.0201
2	1440	21.59	0.02088	3.307	99.9678	-0.0202
3	1520	24.23	0.02416	3.120	99.9668	-0.0208
4	1600	26.87	0.01655	3.907	99.9795	+0.1498
5	1680	29.50	0.01896	5.235	99.9786	-0.0143
6	1760	32.14	0.02667	6.680	99.9724	-0.0253

Case 4 is the largest trend mismatch: the project predicts 99.9795% at 26.87 m/s, whereas the nearest digitised Purnanto simulation point is approximately 99.8297%. Agreement in magnitude for the other five points does not establish validation because the reference trend is not recovered and the current quality definition uses prescribed steam flow with escaped DPM liquid.

3. Size-resolved DPM behaviour

Across all six spiral cases, completed outlet escape occurred only in the finest injection. The 5 µm-class tracks remain closely coupled to the rotating carrier flow, populate the upper recirculation region and exhibit the longest plotted residence-time scale. Increasing diameter shortens the displayed residence-time range and shifts the dominant fate toward bottom trapping, consistent with increasing inertial separation.

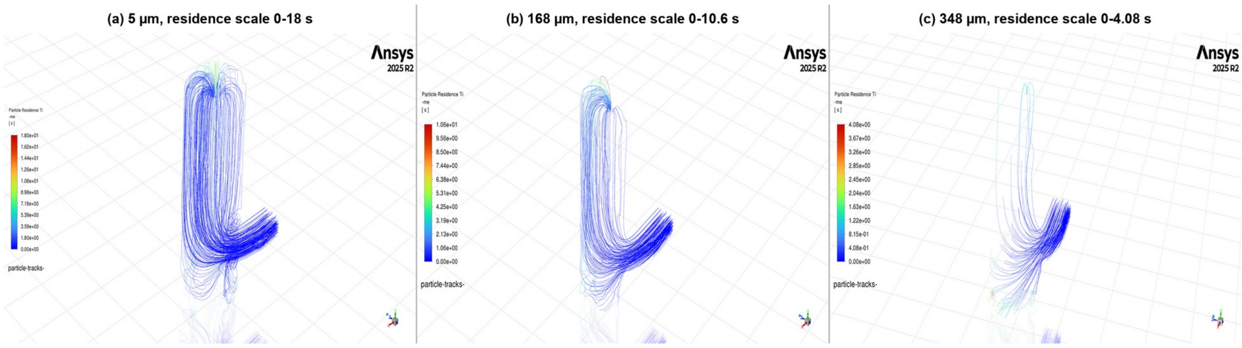

Figure 2. Particle-residence-time tracks for nominal 5, 168 and 348 µm injections. Colour scales differ between panels and therefore indicate within-panel residence time, not a shared quantitative scale.

Table 3. Selected Case 4 (1600 kJ/kg) injection-level fate accounting.

Nominal class (µm)	Actual d (µm)	Injected (kg/s)	Escaped (kg/s)	Trapped (kg/s)	Incomplete (kg/s)	Incomplete (%)
5	5.6	0.1900	0.01655	0.0373	0.1362	71.7
168	168.6	1.9526	0.00000	1.1860	0.7670	39.3
348	348.5	23.3840	0.00000	22.4300	0.9562	4.1

The Case 4 finest bin is an important numerical exception: 0.01655 kg/s escaped, but 0.1362 kg/s remained incomplete, so incomplete mass was over eight times the escaped mass for that bin. No completed escape was recorded for the 168 or 348 µm classes. The 348 µm class instead delivered 22.43 kg/s to the bottom trap, while 0.9562 kg/s remained incomplete. These unresolved tracks are retained explicitly and are not counted as either capture or carryover.

4. Eulerian wall-film mechanism screening

The wall-film family partitioned the liquid input between Eulerian inlet liquid and DPM parcels before enabling deposition and drainage. Mechanisms were then introduced in controlled branches: splash, edge separation, stripping, and a combined model. This avoided double counting the DPM allocation as additional liquid mass.

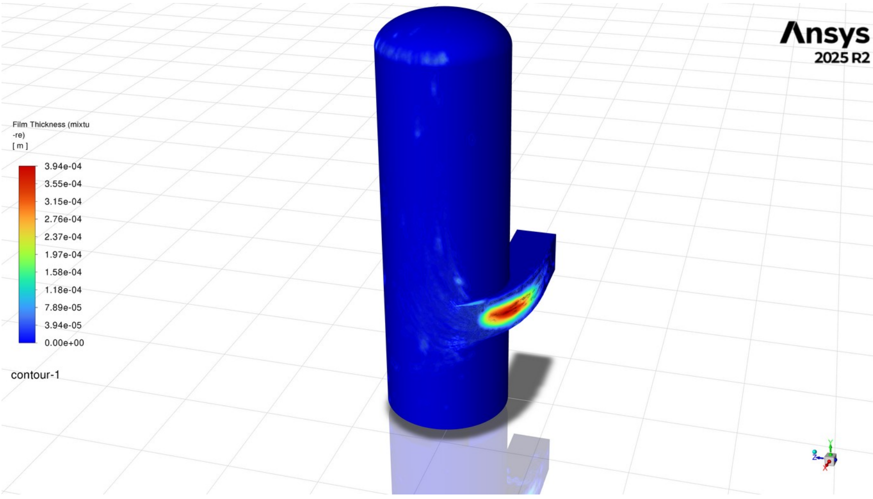

Figure 3. Wall-film thickness on the separator and spiral inlet. The plotted scale spans 0-3.94×10^{-4} m and shows the largest local accumulation at the inlet bend and first vessel-wall impact region.

Table 4. Final film quantities from the mechanism-screening branches.

EWF branch	Inventory (kg)	Maximum (mm)	Area average (µm)
Clean deposition	0.07150	0.152	1.211
Splash only	0.07431	0.164	1.259
Edge separation only	0.05639	0.123	0.955
Stripping only	0.05440	0.121	0.921
Combined mechanisms	0.05668	0.125	0.960

Relative to the clean deposition control, splash increased inventory by 3.9% and maximum thickness by 7.9%, whereas edge separation and stripping reduced inventory by 21.1% and 23.9%, respectively. In the combined EWF branch, enabling global DPM interaction increased inventory, maximum thickness and area-average thickness by 41%, and increased derived mean film speed from 0.0533 to 0.0684 m/s (+28%). The interaction-on state also increased coarse-class splash events from 20 to 44, edge-separation events from 120 to 188, and introduced seven stripping events.

Event counts are not equivalent to re-entrained liquid mass. They demonstrate that the selected wall model changes the assigned liquid pathway, but a quantitative carryover claim requires the generated secondary parcels to be mass-accounted and compared from matched initial film states and physical times.

5. Technical interpretation and limitations

5.1. Iteration completion versus convergence

Every spiral case contains 1500 residual-history points and passed the workflow's setup, save and DPM mass-closure checks. However, fixed iteration count is not convergence evidence. Final continuity residuals remain between 0.145 and 0.229, while momentum residuals are approximately 1.1×10^{-5} to 1.7×10^{-5}. The carrier fields should therefore be described as iteration-complete but not strictly converged.

Table 5. Final scaled residuals after 1500 iterations.

Case	Continuity	Liquid volume fraction	Maximum momentum
1	0.185	8.40e-04	1.33e-05
2	0.229	9.51e-04	1.66e-05
3	0.204	8.90e-04	1.45e-05
4	0.200	8.70e-04	1.50e-05
5	0.162	8.38e-04	1.19e-05
6	0.145	8.57e-04	1.19e-05

5.2. Dominant uncertainty in the particle interpretation

Incomplete liquid increases from 2.4% to 6.6% across the six spiral cases and is substantially larger than escaped liquid. Purnanto also reported incomplete trajectories, so the phenomenon is not unique to the present reconstruction; nevertheless, incomplete particles cannot be reclassified as captured or escaped without spatial evidence. The residence-time plots support localisation of long-lived fine droplets, but a position-resolved export is still required to quantify termination near the cylinder-to-dome transition.

5.3. Evidence boundary

The completed sweep establishes a reproducible numerical pipeline and a size-selective carryover mechanism, but it does not validate separator efficiency. The exact nine-bin mass allocation is reconstructed because Purnanto did not publish the original injection table; the model is isothermal and omits flashing, breakup and coalescence; steam quality uses prescribed steam flow rather than an independently audited outlet vapour monitor; and the reference trend is not recovered. The most defensible mid-year result is therefore that predicted carryover is highly sensitive to droplet size and wall treatment, while the remaining convergence and incomplete-trajectory errors are large enough to control the comparison with published steam quality.

5.4. Technical verification required before final validation

The next numerical checks are to extend the carrier solutions while monitoring integral outlet mass flow and pressure drop, export incomplete-particle endpoints, repeat DPM tracking across step-length and stochastic-dispersion settings, audit quality against Fluent outlet vapour flow, and rerun EWF interaction-off/on cases from a common film state. These checks directly test whether the observed near-constant quality and wall-film changes are physical responses or consequences of unresolved numerical treatment.

Technical reference: Purnanto, Zarrouk & Cater (2013), CFD Modelling of Two-Phase Flow inside Geothermal Steam-Water Separators.